

Proportion of variance explained (PVE)

PVE is how you turn raw eigenvalues into a meaningful answer to the question 'how much information did I keep?'

The **PVE** helps us understand how much variation in the data is explained by specific factors.

For example, in predicting test scores, the proportion of variance *shows how much the difference in the test scores can be attributed to factors like study time, sleep habits, or practice.*

A high proportion indicates that the factor explains most of the variation, whereas a low proportion means it explains little.

The **proportion of variance** is crucial because it:

- Helps evaluate the effectiveness of statistical models in explaining data variability.
- Guides decision-making by highlighting which factors contribute most to variance.
- Supports dimensionality reduction, like PCA, to focus on the most important features, enhancing model efficiency and performance.

Total Variance

Total variance represents the overall spread of the data from the mean and consists of two components: **explained variance (explained by the model)** and **unexplained variance**. The proportion of variance is the fraction of the total variance explained by the model, calculated by:

$$\text{proportion of variance} = \frac{\text{explained variance}}{\text{total variance}}$$